SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE
Constituent of Symbiosis International (Deemed University), Pune

Course	Digital Signal Processing Lab	**Experiment No.**	01
Name	Dewashish Lambore	**PRN**	_____
Language	Python 3.10	**Date**	____ / ____ / 2026

Basic Operations on Discrete-Time Signals

Applied to hourly solar-irradiance and weather signals of Pune

Abstract: One week (N = 168 hourly samples) of real ERA5 reanalysis data for Pune was treated as a set of discrete-time signals, and nine elementary operations were implemented in a modular, tested Python library with an interactive GUI. Each operation solved a physical task: addition reconstructed global irradiance with zero error, subtraction flagged 11 fog-risk hours, multiplication estimated a PV yield of 5.29 kWh/kWp/day (9.9 % thermal loss), a 24-sample delay gave a persistence forecast with 74 % skill, and a 24-point moving-average convolution cut the standard deviation from 3.60 °C to 0.30 °C, exposing the multi-day trend.

Keywords: *discrete-time signals, convolution, moving average, sampling, solar energy, Python*

1. Aim and Objectives

Aim: to perform the basic operations on discrete-time signals (addition, subtraction, multiplication, amplitude scaling, time shifting, time reversal, time scaling and convolution) on a real-world dataset in Python, and to interpret each result physically.

- Represent measured data as sequences $x[n]$ with an explicit index support.
- Implement every operation as a reusable, unit-tested function and expose it through a GUI.
- Verify theory numerically (energy scaling, convolution length) and relate each operation to a real engineering task.

2. Real-World Application

Rooftop-solar capacity in Maharashtra is growing quickly, and every inverter, weather station and energy-management system in such an installation works on sampled data. The operations of this experiment are the first stage of those monitoring pipelines (Table 1): pyranometer components are added, dew point is subtracted from temperature to warn of condensation on panels, irradiance is multiplied by a thermal factor to predict power, yesterday's profile is shifted to forecast today, loggers down-sample to save bandwidth, and moving averages expose trends.

Table 1. Operations, their use here and in industry

Operation	Definition	Use in this experiment	Industrial application
Addition	$x_1[n] + x_2[n]$	Direct + diffuse = global irradiance	Sensor fusion, station QC
Subtraction	$x_1[n] - x_2[n]$	Temperature − dew point (fog indicator)	Noise / baseline removal
Multiplication	$x_1[n]\, x_2[n]$	Irradiance × thermal factor = PV power	Power metering, windowing
Amplitude scaling	$A\, x[n]$	1 kWp → 3 kWp array	Calibration, gain control
Time shift	$x[n-k]$	24-h delay = persistence forecast	Latency compensation
Time reversal	$x[-n]$	Folding, even/odd parts	Matched / zero-phase filtering
Down-sampling	$x[Dn]$	Hourly → 3-hourly synoptic data	IoT data compression
Up-sampling	$x[n/L]$	3-hourly → hourly reconstruction	Sample-rate conversion
Convolution	$x[n] * h[n]$	24-h moving average (trend)	Smoothing, FIR filtering

3. Theory

A discrete-time signal is a sequence $x[n]$ defined only at integer n. Sampling a continuous signal every T_s seconds gives $x[n] = x_a(nT_s)$; here T_s = 1 h, so $n = 0$ is 00:00 IST on 1 October 2026. A finite signal is non-zero only on its **support** $n_1 \le n \le n_2$, and its energy and average power are

$$E_x = \sum_{n=n_1}^{n_2} |x[n]|^2, \qquad P_x = \frac{E_x}{N} \tag{1}$$

Arithmetic operations act sample-by-sample; operations on the index n change *where* the signal lives. Table 2 gives the effect of each on support, length and energy.

Table 2. Effect of each operation on support, length and energy

Operation	Output support	Length	Energy	Remark
$A\, x[n]$	$[n_1, n_2]$	N	$A^2 E_x$	amplifies (|A|>1) or attenuates
$x[n-k]$	$[n_1 + k, n_2 + k]$	N	E_x	k > 0 delay, k < 0 advance
$x[-n]$	$[-n_2, -n_1]$	N	E_x	mirror about n = 0
$x[Dn]$	$[n_1/D, n_2/D]$	$\approx N/D$	$\approx E_x/D$	keeps every D-th sample; may alias
$x[n/L]$	$[Ln_1, Ln_2]$	$L(N-1)+1$	E_x	inserts L − 1 zeros; needs interpolation
$x[n] * h[n]$	starts at $n_x + n_h$	$N + M - 1$	depends on h	LTI filtering (Eq. 2)

Convolution gives the output of a linear time-invariant system with impulse response $h[n]$. The causal M-point moving average has $h[n] = 1/M$ for $0 \le n \le M-1$:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k] = \frac{1}{M}\sum_{k=0}^{M-1} x[n-k] \tag{2}$$

Its DC gain $\sum h[n] = 1$ passes constants unchanged, while any oscillation whose period divides M is cancelled; with $M = 24$ the daily cycle is removed. The output lags by $(M-1)/2$ samples.

Application models. Global irradiance is the sum of beam and diffuse components; PV power is irradiance multiplied by a thermal derating factor (PVWatts model, NOCT cell temperature); and a forecast is scored by RMSE and skill against the weekly mean:

$$\mathrm{GHI}[n] = \mathrm{DNI}[n]\cos\theta_z[n] + \mathrm{DHI}[n] \tag{3}$$

$$P[n] = \frac{G[n]}{1000} \cdot [1 + \gamma(T_a[n] + \frac{\mathrm{NOCT}-20}{800} G[n] - 25)] \tag{4}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N'}\sum(x[n] - \hat{x}[n])^2}, \qquad S = 1 - \frac{\mathrm{RMSE}}{\mathrm{RMSE}_{\mathrm{clim}}} \tag{5}$$

Note. with γ = -0.0037 /°C and NOCT = 45 °C. Signals are stored as pairs (n, x) in the code: padding with zeros and keeping n = 0 … N−1 (a common shortcut) loses samples on shifting and makes x[−n] indistinguishable from simply reversing an array.

4. Dataset Description

Hourly **ERA5 / ERA5-Land reanalysis** data (ECMWF, Copernicus Climate Change Service) for Pune (18.5204° N, 73.8567° E, 561 m), from 2026-10-01 to 2026-10-07 (the post-monsoon transition), downloaded from the free Open-Meteo archive API. Reanalysis blends satellite, radiosonde and ground observations with a physical model, so the record is gap-free and physically consistent. The file is only 8.7 kB (N = 168 samples × 7 signals, no missing values) and is cached locally after the first run.

Table 3. Dataset attributes and statistics (168 hourly samples)

Column	Signal	Unit	Mean	Min	Max	Role
`temp_c`	Air temperature	°C	27.5	21.9	33.7	x_1 (subtraction, shift, scaling, conv.)
`dewpoint_c`	Dew-point temperature	°C	17.6	10.2	22.2	x_2 (subtraction)
`rh_pct`	Relative humidity	%	58.9	25.0	95.0	check for DPD
`ghi_wm2`	Global horizontal irradiance	W/m²	244.6	0.0	896.0	reference; PV input
`direct_wm2`	Direct horizontal irradiance	W/m²	176.4	0.0	718.0	x_1 (addition)
`diffuse_wm2`	Diffuse horizontal irradiance	W/m²	68.2	0.0	266.0	x_2 (addition)
`cloud_pct`	Cloud cover	%	21.6	0.0	99.0	context

Table 4. Dataset preview around sunrise, 1 October (irradiance in W/m²; GHI = Direct + Diffuse)

n	Time (IST)	T (°C)	Td (°C)	RH (%)	GHI	Direct	Diffuse	Cloud (%)
6	01-Oct 06:00	22.3	20.9	92	0	0	0	20
7	01-Oct 07:00	24.0	21.0	83	68	28	40	3
8	01-Oct 08:00	26.6	20.4	69	273	171	102	2
9	01-Oct 09:00	28.4	19.0	57	495	360	135	3
10	01-Oct 10:00	29.8	18.3	50	684	525	159	3
11	01-Oct 11:00	31.3	16.4	41	826	655	171	12

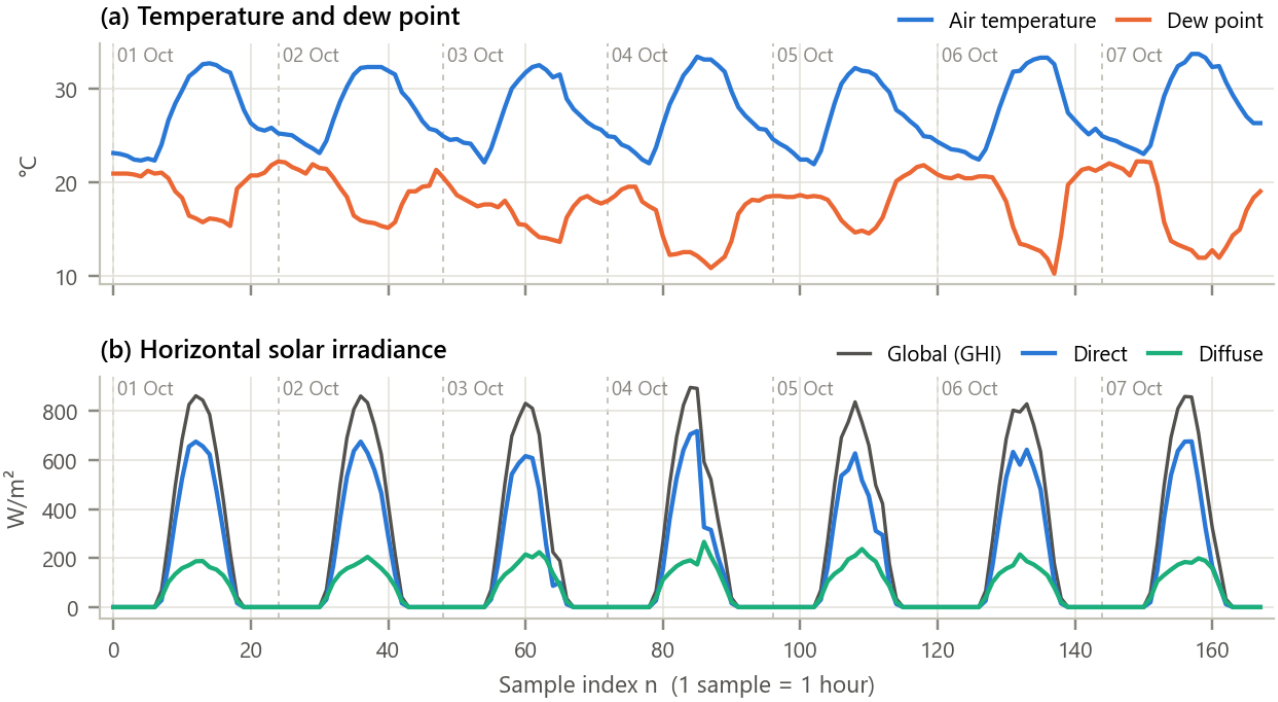


Figure 1. Raw input signals for the full week. Dashed lines mark midnight.

> **Inference.** Every signal is dominated by a 24-sample (daily) periodicity. Temperature swings by ≈ 10.5 °C per day while dew point stays flatter; irradiance is zero for ≈ 12 h each night and peaks at 830 to 900 W/m², indicating mostly clear skies with brief afternoon cloud on 3 and 4 October. Because the signals are physically linked, arithmetic between them has real meaning.

5. Methodology and Implementation

The code is split into single-responsibility modules (Table 5). Each column is wrapped in a `DTSignal(n, x)` object that carries its own index vector, so a delay adds k to n and a fold negates n exactly as in Table 2. Parameters: k = 24, A = 3, D = L = 3, M = 24 (and 3), fog threshold 2 °C.

Table 5. Software modules

Module	Responsibility
`config.py`	All constants, paths and experiment parameters
`data_loader.py`	Download from Open-Meteo once, cache as CSV, clean, attach index n
`signal_ops.py`	`DTSignal` class and the nine pure operations (no I/O)
`analysis.py`	Physical interpretation: GHI closure, DPD, PV power, forecast skill, smoothing
`plots.py` / `gui.py`	Report figures / interactive Tkinter application
`test_signal_ops.py`	20 unit tests of the identities in Table 2 (all pass)

Core of signal_ops.py

```python
@dataclass
class DTSignal:
    n: np.ndarray                    # integer sample indices
    x: np.ndarray                    # sample values x[n]

def time_shift(s, k):                # y[n] = x[n - k]
    return DTSignal(s.n + k, s.x.copy())

def time_reverse(s):                 # y[n] = x[-n]
    return DTSignal(-s.n[::-1], s.x[::-1].copy())

def downsample(s, D):                # y[n] = x[Dn]
    keep = s.n % D == 0
    return DTSignal(s.n[keep] // D, s.x[keep].copy())

def convolve(s, h):                  # support starts at n_x + n_h, length N + M - 1
    y = np.convolve(s.x, h.x)
    return DTSignal(np.arange(len(y)) + s.n[0] + h.n[0], y)

```

GUI. `gui.py` (Figure 2) lets the user pick $x_1[n]$ and $x_2[n]$ from the dataset, choose any of the nine operations, and set its parameter (A, k, D, L or M) with a slider; the equation, the plots and a statistics table (N, support, mean, min, max, energy, power) update instantly. Run with `python run_experiment.py` (all figures and results) and `python gui.py`.

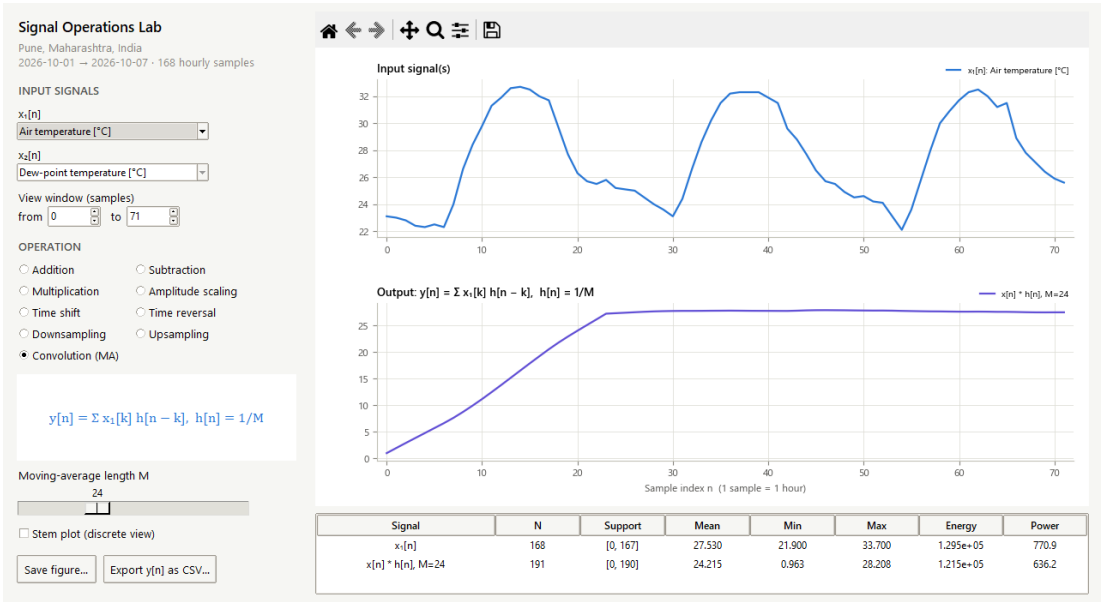


Figure 2. Interactive GUI applying a 24-point moving average. The table reports an output length of 191 = 168 + 24 − 1, as predicted by Table 2.

6. Calculations

Using the first four samples, $x_1[n] = \{23.1, 23.0, 22.8, 22.4\}$ (temperature) and $x_2[n] = \{20.9, 20.9, 20.9, 20.8\}$ (dew point), $n = 0 \ldots 3$:

Table 6. Hand calculation on four samples

Operation	Working	Result	Support
$x_1 + x_2$	23.1+20.9 , 23.0+20.9 , 22.8+20.9 , 22.4+20.8	44.0, 43.9, 43.7, 43.2	[0, 3]
$x_1 - x_2$	23.1−20.9 , 23.0−20.9 , 22.8−20.9 , 22.4−20.8	2.2, 2.1, 1.9, 1.6	[0, 3]
$3 \cdot x_1$	3 × each sample	69.3, 69.0, 68.4, 67.2	[0, 3]
$x_1[n - 2]$	same values, $n \rightarrow n + 2$	23.1, 23.0, 22.8, 22.4	[2, 5]
$x_1[-n]$	order reversed, $n \rightarrow -n$	22.4, 22.8, 23.0, 23.1	[−3, 0]
$x_1[2n]$	keep n = 0, 2	23.1, 22.8	[0, 1]
$x_1[n/2]$	zero between samples	23.1, 0, 23.0, 0, 22.8, 0, 22.4	[0, 6]
x_1 * h, h = {⅓, ⅓, ⅓}	$y[n] = (x[n] + x[n-1] + x[n-2])/3$	7.700, 15.367, 22.967, 22.733, 15.067, 7.467	[0, 5]

For the convolution, e.g. $y[2] = (23.1 + 23.0 + 22.8)/3 = 22.9667$; the output length is $4 + 3 - 1 = 6$, and y[0], y[1], y[4], y[5] are partial-overlap edge values. **Energy:** $E(x_1) = 23.1^2 + 23.0^2 + 22.8^2 + 22.4^2 = 2084.21$ °C², so $E(3x_1) = 9 \times 2084.21 = 18757.89$ °C². **PV at 09:00:** $T_c = 28.4 + 0.03125 \times 495 = 43.87$ °C, $f_T = 1 - 0.0037 \times 18.87 = 0.9302$, $P = 0.495 \times 0.9302 =$ **0.4604 kW**. All values agree with the program output to four decimals (checked automatically by `test_signal_ops.py`).

7. Results and Discussion

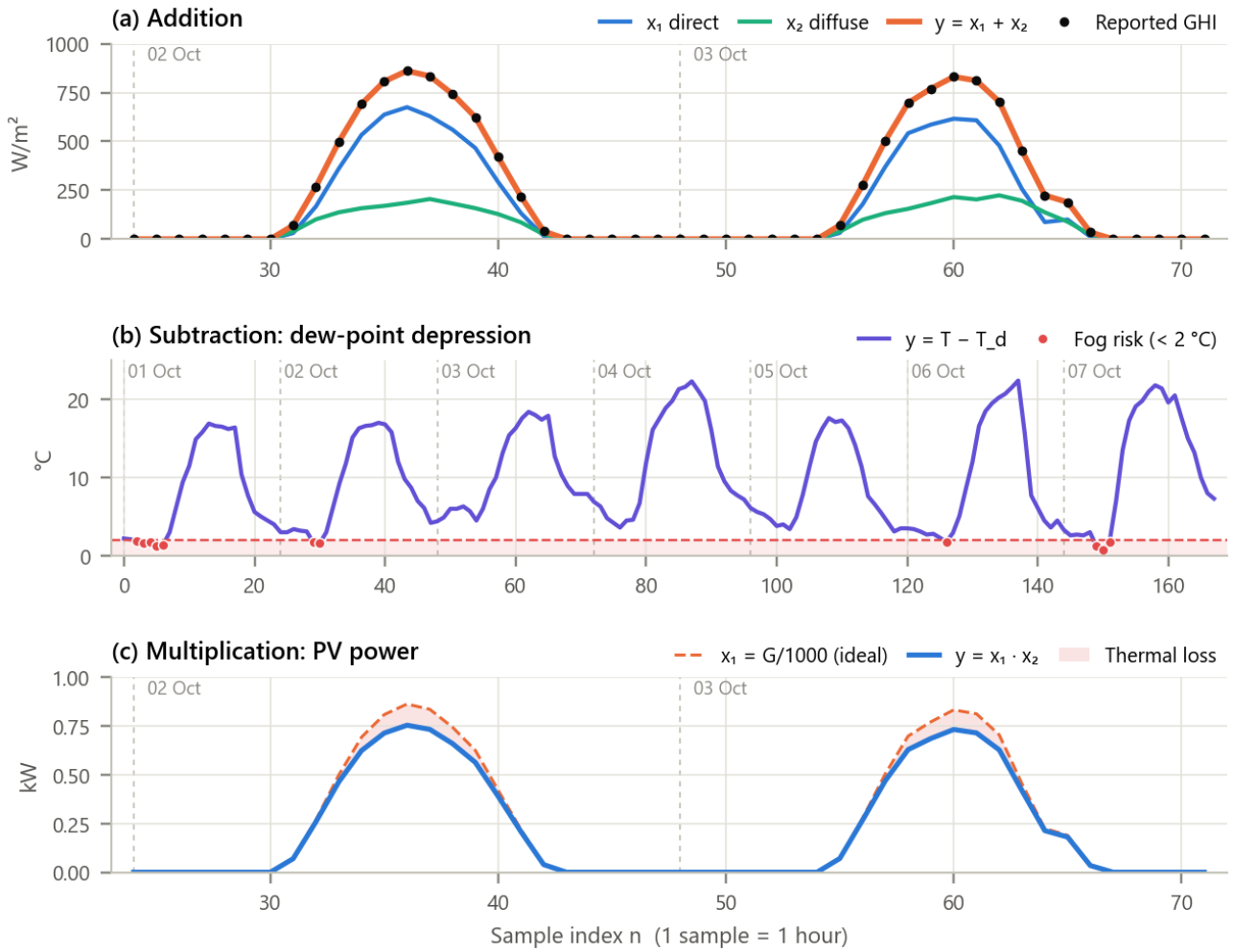


Figure 3. Arithmetic operations on real signals. (a) Direct + diffuse irradiance vs. the independently reported GHI; (b) temperature − dew point with the fog-risk band; (c) normalised irradiance × thermal factor = PV power (2 and 3 October).

Inference.

- **Addition** reproduces the reported GHI at every sample (maximum residual 0.0 W/m²), so the operation is exact; diffuse light supplies 27.9 % of daylight energy, typical of clear tropical skies.
- **Subtraction** gives a dew-point depression of 0.8 to 22.4 °C. It drops below 2 °C in 11 hours, all between 02:00 and 07:00 (when dew forms on panels), and correlates with humidity at r = -0.990, i.e. two temperatures give a humidity indicator.
- **Multiplication** cuts output most exactly when irradiance is highest: cells reach 61.3 °C and the factor falls to 0.866. The 1 kWp array yields 5.03 to 5.54 kWh/day, losing 9.9 % to heat.
- **Amplitude scaling** by A = 3 (3 kWp) keeps shape and timing; the kWh ratio is 3.00 = A while the energy ratio $\Sigma y^2/\Sigma x^2$ is 9.00 = A^2, confirming Table 2.

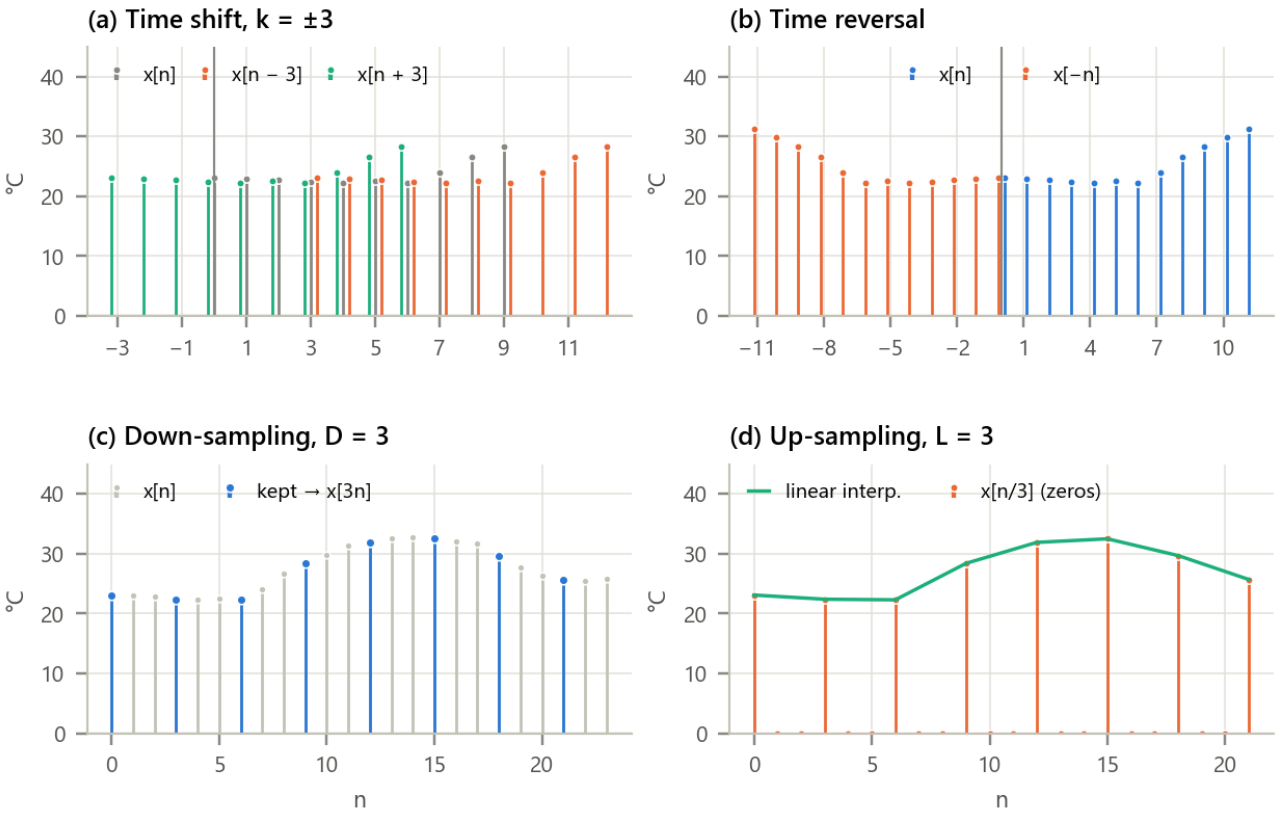


Figure 4. Operations on the independent variable n. (a) Delay and advance by 3 on ten samples; (b) folding of twelve samples; (c) down-sampling by 3; (d) zero-insertion up-sampling by 3 with linear interpolation.

Inference.

- Shifting and folding change only the **position** of samples: the delay moves the support to [3, 12], the advance to [−3, 6], the fold to [−11, 0]. No sample is lost because n moves with x. Splitting into even and odd parts and adding them back reproduces x[n] exactly (error 0.0).
- Down-sampling keeps 56 of 168 samples and 33.1 % of the energy (≈ 1/D). Up-sampling gives 166 = 3(56 − 1) + 1 samples, two-thirds zeros; linear interpolation restores the hourly curve with RMSE 0.42 °C, but the 3-hourly record under-reads the daily maximum by 0.20 °C on average, a sampling (aliasing) bias.

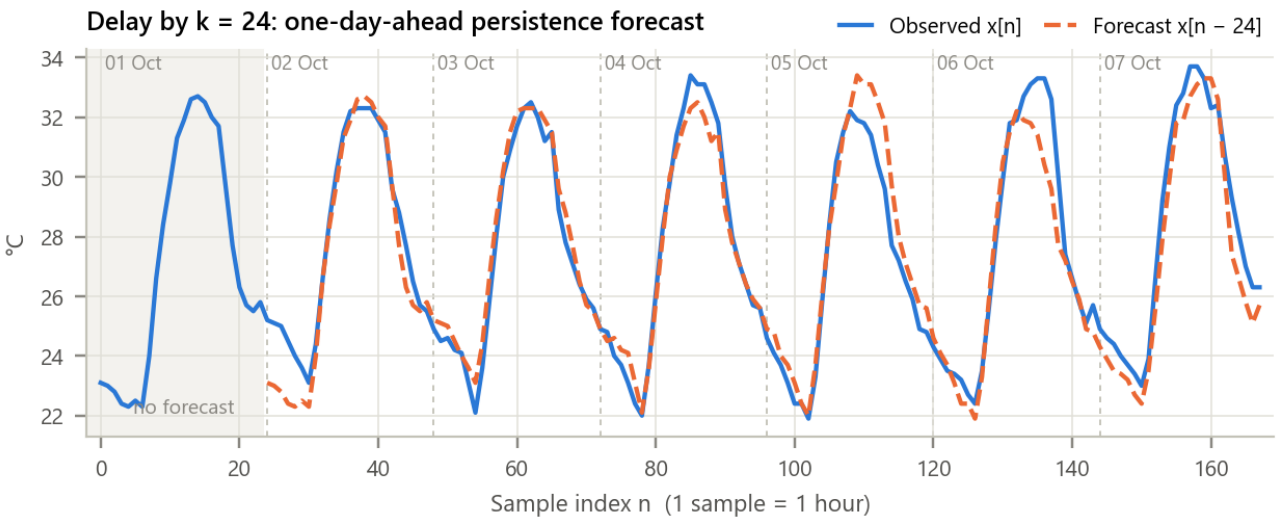


Figure 5. Time delay as a forecast. The observed temperature and the same signal delayed by one day.

Inference. A delay of k = 24 samples is the meteorologist's persistence forecast. It achieves MAE 0.67 °C and RMSE 0.91 °C, against 3.56 °C for predicting the weekly mean, a skill of 74.4 %. Errors grow when consecutive days differ (e.g. the cooler afternoon of 5 October); any real forecast must beat this baseline.

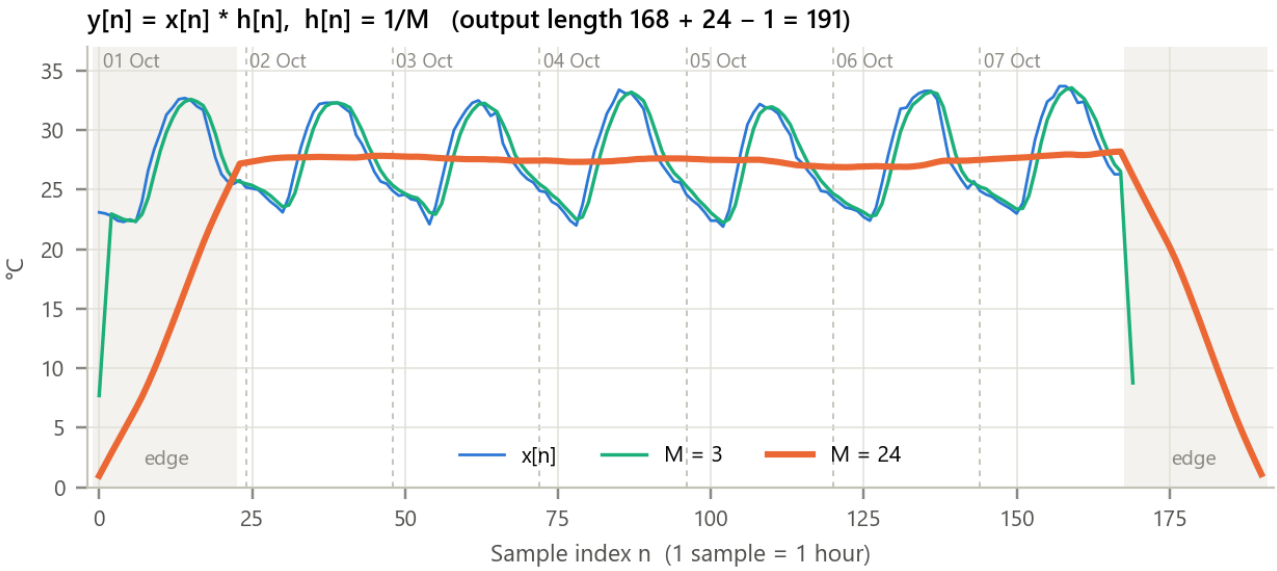


Figure 6. Convolution with moving-average kernels (M = 3 and M = 24). Shaded regions are the partial-overlap edges.

Inference. The output length is 191 = 168 + 24 − 1, as predicted. M = 3 only lightly smooths; M = 24 removes the daily cycle completely, lowering the standard deviation from 3.60 °C to 0.30 °C (92 % reduction) and revealing the daily-mean trend between 26.9 and 28.2 °C. The ramps at both ends are edge transients and the output lags by (M − 1)/2 = 11.5 h; both matter when moving averages are used for monitoring.

Table 7. Summary of results

Operation	Output support	Length	Key result
Addition	[0, 167]	168	GHI residual 0.0 W/m²
Subtraction	[0, 167]	168	11 fog-risk hours; r = -0.990
Multiplication	[0, 167]	168	5.29 kWh/kWp/day; 9.9 % loss
Amplitude scaling	[0, 167]	168	energy × 9.00 = A²
Time shift (k = 24)	[24, 191]	168	RMSE 0.91 °C; skill 74.4 %
Down-sampling (D = 3)	[0, 55]	56	energy 33.1 %
Up-sampling (L = 3)	[0, 165]	166	reconstruction RMSE 0.42 °C
Convolution (M = 24)	[0, 190]	191	std 3.60 → 0.30 °C

8. Conclusion

All nine basic operations were implemented and applied to real weather and solar data. Wherever theory gives an exact answer the results matched it (zero GHI residual, energy ratio A^2, output length $N + M - 1$, perfect even/odd reconstruction), and elsewhere they produced useful engineering quantities: a fog-risk window, a 9.9 % PV thermal loss, a 74 %-skilful forecast from a simple delay, and a clean trend from a moving average. Two lessons stand out: index operations are only correct if the support is tracked, and each operation has practical side-effects (sampling bias, interpolation error, edge transients and group delay) that must be accounted for in real systems.

9. Learning Outcomes

- Represent sampled data as discrete-time signals and predict the support, length and energy of the output of any basic operation.
- Implement the operations from first principles in modular, unit-tested Python with a GUI.
- Connect each operation to a real task: sensor fusion, condensation warning, PV estimation, forecasting, data compression and trend extraction.
- Recognise aliasing, interpolation error, edge effects and group delay in practical results.

10. References

[1] A. V. Oppenheim and R. W. Schafer, *Discrete-Time Signal Processing*, 3rd ed. Pearson, 2010.
[2] J. G. Proakis and D. G. Manolakis, *Digital Signal Processing*, 4th ed. Pearson, 2007.
[3] H. Hersbach et al., "The ERA5 global reanalysis," *Q. J. R. Meteorol. Soc.*, vol. 146, pp. 1999-2049, 2020.
[4] P. Zippenfenig, "Open-Meteo.com Weather API," Zenodo, 2023, doi:10.5281/zenodo.7970649.
[5] A. P. Dobos, "PVWatts Version 5 Manual," NREL Tech. Rep. NREL/TP-6A20-62641, 2014.